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Palaeomagnetism of the Fongen-Hyllingen gabbro complex, southern Scandinavian Caledonides; plate rotation or polar shift?

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Summary. Palaeomagnetic results are presented from the *c.* 160 km^2 Caledonian synorogenic layered Fongen–Hyllingen gabbro complex (of probable late Silurian age) located about 75 km SE of Trondheim, Norway, in the allochthonous Seve-Køli Nappe Complex. A total of 80 oriented samples from eight sites in the northern part of the gabbro were investigated. After detailed af demagnetization two stable high coercivity components emerge: one with a well defined NW direction with $D = 325°$, $I = -21°$ ($\alpha_{95} = 8°$, $N = 8$), and another, less well defined, probably younger, SW direction with $D = 237°$, $I = 6°$ ($\alpha_{95} = 9°$, $N = 8$). Correction for dip of these two directions gives $D = 329°$, $I = -7°$ ($\alpha_{95} = 10°$) and $D = 238°$, $I = -11°$ ($\alpha_{95} = 12°$), respectively. The corresponding pole positions are P_1: 19° N, 225° E and P_2: 19° S, 308° E, respectively. The reversed pole $-P_2$ of the SW direction lies close to other NW European palaeomagnetic poles of Caledonian, Upper Silurian–Lower Devonian age. However, the dominant pole P_1 is far away from these, and could be due to a late Caledonian geomagnetic excursion of considerable duration; or it could record a *c.* 90° rotation around a vertical axis of a crustal block within the Scandinavian Caledonides. Block rotation could have been related to nappe translation, although geological observations do not at present appear to support the occurrence of such an event.

1 Introduction

The Fongen–Hyllingen complex is a synorogenic layered gabbro intrusion, situated about 75 km south-east of Trondheim in the Norwegian Caledonides (Fig. 1). A preliminary account of the complex has been presented by Wilson & Olesen (1975). The present study has been undertaken to add to the scanty palaeomagnetic results from the Scandinavian Caledonides.

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2 Geology

2.1 REGIONAL SETTING OF THE FONGEN–HYLLINGEN COMPLEX

The Fongen–Hyllingen complex is emplaced in rocks belonging to the allochthonous Western Complex (e.g. Gee 1975) of the southern Scandinavian Caledonides. The gabbro intrudes metasediments and metavolcanics of the Fundsjø group, which may correlate with the Støren group to the west (e.g. Roberts, Springer & Wolff 1970), and the Ordovician (Tremadocian) fossil Dictyonema flabelliforme has been found in rocks of the aureole (Vogt 1940). The gabbro hornfelsed tightly folded rocks and has inclusions of folded country rocks, so the intrusion clearly post-dates at least one fold phase. A contact metamorphic aureole with index-minerals cordierite, sillimanite and andalusite is developed, with additional garnet present throughout. The peak of regional metamorphism was reached shortly after the emplacement of the complex and was accompanied by the main deformation phase in the area (Olesen *et al.* 1973). In the western part of the aureole the contact metamorphic assemblages are overprinted by regional assemblages such as andalusite being replaced by kyanite and sillimanite by staurolite and white mica.

During regional metamorphism, large parts of the Fongen–Hyllingen complex were altered to hornblende metagabbros, though often retaining igneous microstructures, and foliated amphibolites were produced where metamorphism of the gabbro was accompanied by penetrative deformation. The gabbro and its envelope were later folded during the Selbu–Tydal phase (Olesen *et al.* 1973) and the present form of the Fongen–Hyllingen complex (Wilson & Olesen 1975) is to some extent influenced by its situation in the Tydal synform which is overturned towards the east. The gabbro can thus with certainty be regarded as synorogenic.

The Fundsjø Group rocks enveloping the complex occupy a controversial tectonostratigraphic position but are interpreted by Zwart (1974) and Gee (1975) as part of the Køli (Supergroup) of the Seve–Køli Nappe complex. Orogenic deformation of this nappe, with folding, metamorphism and thrusting is considered by Gee & Wilson (1974) to have both started and finished in Silurian times. Important evidence as to the timing of the end of the Caledonian orogeny in this region is found at Røragen, some 50 km SSE of Fongen, where rocks probably belonging to the Køli Supergroup are unconformably overlain by continental conglomerates and sandstones, identified from plant fossils as of Lower Devonian age (Halle 1916; Høeg 1936). These Devonian rocks have however been affected by a late phase of the Caledonian orogeny since they are folded, faulted and show evidence of low grade metamorphism (Holmsen 1963), and the possibility remains that they were transported along with the pre-Devonian rocks of the Seve–Køli Nappe during the final thrusting.

2.2 PETROLOGY OF THE COMPLEX

Rocks of the Fongen–Hyllingen complex vary from ultramafics, through olivine gabbros, gabbros and norites to diorites, monzonites and quartz-bearing syenites. Rhythmic layering is widely developed and there is an extensive cryptic variation (Esbensen, Thy & Wilson 1978). The deepest levels of the intrusion occur in the north where layering is arranged in a saucer-like form, emphasized by later folding. In the southern part of the complex around Hyllingen mountain, described by Kisch (1962) and Nilsen (1973), the rhythmic layering strikes roughly north–south and dips east, the rocks younging consistently to the east.

On the basis of the phase layering and cryptic variation, the complex can be divided into three parts. These are the Fongen and Ruten sequences in the north (Fig. 1) and the

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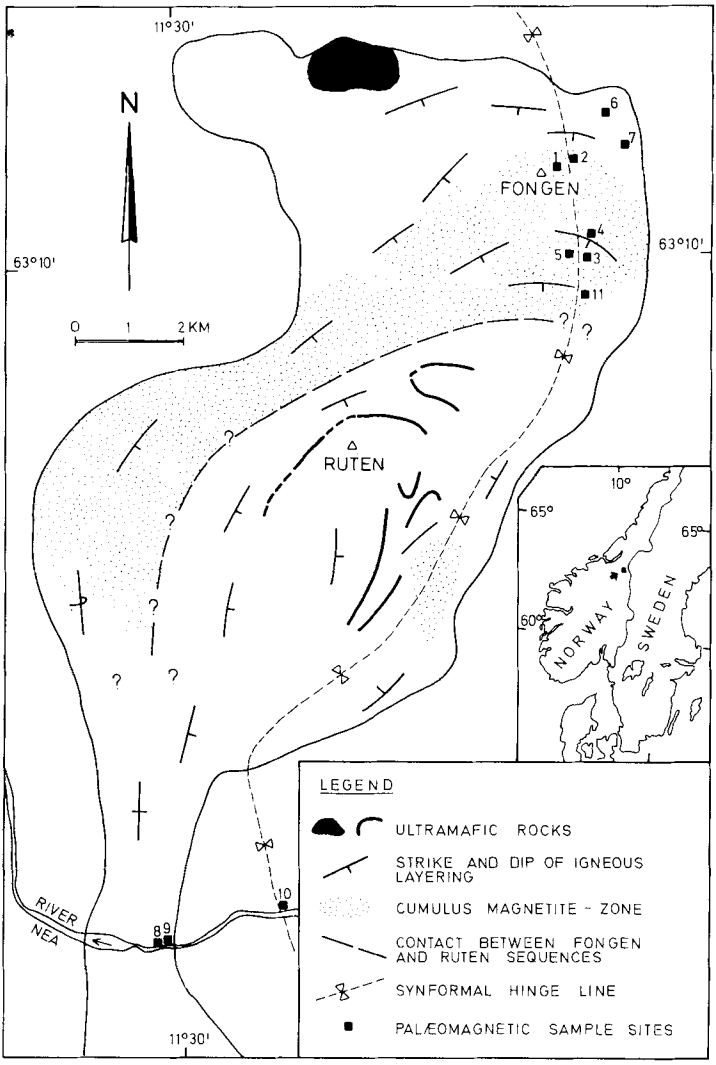


Figure 1. Simplified geological map of the northern part of the Fongen–Hyllingen gabbro complex; Nos 1–11 show the sites of palaeomagnetic sampling.

Hyllingen sequence to the south (Esbensen *et al.* 1978). Each consists of a series of cumulates about 4000 m, 1800 m and 3000 m thick, respectively.

In the Fongen sequence the lowest cumulates consist of olivine (Fo_{87}) accompanied by chromite. Plagioclase joins olivine roughly when chromite ceases to occur, and these are accompanied by Ca-rich and Ca-poor pyroxenes higher up. Fe–Ti oxides first crystallize as an intercumulus and then as a cumulus phase, and the olivine + plagioclase + Ca-rich and Ca-poor pyroxenes + Fe–Ti oxides cumulus assemblage is joined by apatite in the highest rocks of the Fongen sequence. Cumulus Fe–Ti oxides occur in an average amount of 5 vol. per cent (range 0–10 per cent) throughout a thickness of some 800 m around and to the south of the top of Fongen mountain (Fig. 1). These rocks give rise to a major positive anomaly on the 1:50000 aeromagnetic map of Tydal (Norges Geologiske Undersøkelse, 1721, III). It is from this area that the major part of the palaeomagnetic sampling was

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carried out. The Fongen sequence is overlain by the Ruten sequence in which olivine + plagioclase + Ca-rich pyroxene cumulates at the base are joined upwards by Ca-poor pyroxene and then Fe–Ti oxides. The Ruten sequence is considered to represent the fractionally crystallized products of a new influx of magma after consolidation of the earlier Fongen sequence.

The southern Hyllingen area represents the upper part of a thick, differentiated gabbroic sequence and is separated from the northern Fongen–Ruten area by the river Nea valley where exposures are scanty.

2.3 OPAQUE PHASES FROM THE FONGEN–HYLLINGEN COMPLEX

The primary oxides in the northern part of the complex are titanomagnetite belonging to the magnetite–ulvöspinel solid solution series, and ilmenite belonging to the ilmenite–haematite solid solution series. Titanomagnetite (30 mol per cent ulvöspinel) normally dominates over ilmenite (0–2 mol per cent haematite). Two microstructural varieties can be distinguished; ilmenite and titanomagnetite occurring as intercumulus grains interstitial to silicates, and subhedral to euhedral cumulus grains in which titanomagnetite dominates over ilmenite. The grain size of the cumulus Fe–Ti oxides ranges from 1 mm to below 0.05 mm.

Two types of titanomagnetite are found, the first indicating an early stage of ulvöspinel exsolution and the second a sub-solidus oxidation of ulvöspinel at high temperature during magmatic or late-magmatic conditions (Buddington & Lindsley 1964). Ulvöspinel can be completely or partly replaced by magnetite/ilmenite intergrowth and is found in one third of the samples where it occurs in a fine cloth-like microstructure with possible single or pseudosingle domain magnetite in the micron-range. The most common microstructures are, however, a subhedral coarse ilmenite/magnetite intergrowth or an intergrowth of fine ilmenite lamellae in magnetite corresponding to oxidation class II–III (Ade-Hall *et al.* 1968). Similar assemblages have been described, among others by Vincent (1960) and Mathison (1975). Fine acicular Al–Fe spinel is found in the magnetite and, as proposed by Turnock & Eugster (1962), results from exsolution.

Ilmenite occurs as mm-sized interstitial grains, sometimes with tapering twins. Ilmenite can replace titanomagnetite when they have mutual grain boundaries. Equivalent microstructures have been interpreted as resulting from replacement during crystallization (Gierth & Krause 1973) or as a sub-solidus reaction (Krause & Pape 1975). Haematite exsolution is not observed, which according to Ramdohr (1969) indicates less than 6 per cent Fe_2O_3 in the ilmenite–haematite solid solution series.

Sulphides occur in amounts less than 2 per cent of the total opaque phase in the cumulus magnetite zone. Pyrrhotite is the dominant sulphide and may contribute to the remanence in a few cases. It occurs in interstitial to subhedral aggregates with minor amounts of pentlandite and chalcopyrite with sub-solidus exsolution microstructures.

Secondary magnetite, and to a lesser extent ilmenite, is formed during late-magmatic replacement of olivine with magnetite–orthopyroxene symplectites or by reactions between olivine and plagioclase. The magnetite occurs with the silicates in a vermicular to tabular symplectic intergrowth or as irregular grains from a few microns to mm-size often surrounding partly or completely altered olivine. Magnetite and ilmenite are also found as tabular to acicular grains along the prismatic cleavage in the pyroxenes. Magnetite is only altered to a relatively minor degree with a weak marginal decomposition to goethite in some instances.

The opaque phases hence record a prolonged crystallization and sub-solidus re-equilibration history. The ilmenite/magnetite intergrowth clearly post-dates the ulvöspinel/magnetite

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intergrowth, and is interpreted as a sub-solidus oxidation from ulvöspinel (Buddington & Lindsley 1964) which occurred during or after the metamorphism. The exsolution of Al–Fe spinel post-dates the ulvöspinel exsolution. The oxides derived from silicates are considered to be a synantectic result of magmatic cooling at elevated temperature (Esbensen 1978), as well as a result of late magmatic interactions, as described by Ambler & Ashly (1977). The sulphides show susceptibility to low temperature re-equilibration during or after the oxidation of ulvöspinel to ilmenite.

3 Palaeomagnetic investigations

A total of 111 palaeomagnetic samples were collected from eight sites in the gabbro around Fongen mountain (Fig. 1, sites 1–7 and 11), two sites in strongly deformed and amphibolitized gabbro (sites 8 and 9) and one site in a trondhjemite intrusion (site 10).

In the field 10 hand samples were collected from each site with an areal coverage of typically $30 \times 100\ m^2$. Unmetamorphosed sites were selected from previous knowledge. Unweathered hand samples were loosened by a hammer, the north direction was measured by means of a magnetic compass and marked on a horizontal surface of plaster of Paris. The compass measurements were individually controlled by topographical sights and no significant discrepancies were encountered. The strike and dip of the strata were further measured in 10 places at each site, and the palaeomagnetic site mean directions (Table 1) were also later corrected for the initial dip (see below); the mean errors of the mean dip and strike are estimated to be about $\pm 1^\circ$ and $\pm 2^\circ$ respectively.

In the laboratory several 1-in. oriented cores were drilled from each sample, and a total of 379 1 x 1-in. specimens were thus obtained, 302 being from the gabbro proper. The palaeomagnetic results from the amphibolites (sites 8 and 9) and the trondhjemite (site 10) were extremely scattered, and are not considered further.

The geometrical mean of the volume susceptibility of the gabbro measured partly on hand samples (11) and partly on cored specimens (14) was 2×10^{-3} G/Oe (range $3.0–400 \times 10^{-4}$), and the mean of the respective Q-values was 2 (range 0.4–10). The natural remanent magnetization (NRM) of all the specimens was initially measured using a Digico spinner magnetometer. The direction of the mean NRM of the gabbro was $D = 327^\circ$, $I = 64^\circ$ ($\alpha_{95} = 5.2^\circ$, $N = 302$ specimens), which is fairly close to the present geomagnetic field ($D_0 = 356^\circ$, $I \simeq 75^\circ$) as well as the axial dipole field ($D_0 = 0^\circ$, $I_0 = 75.8^\circ$). A dominating viscous remanent magnetic component is therefore present as a recent overprint. The intensity of the NRM varied 4 orders of magnitude (range 1.6×10^{-5} to 1.4×10^{-1} emu/cc) with the geometrical mean within the sites ranging between 0.3×10^{-3} and 4.5×10^{-3} emu/cc, and with an overall geometrical mean of 2×10^{-3} emu/cc, $N = 302$. After some pilot demagnetizations, one specimen from each hand sample was stepwise af demagnetized in zero field ($\pm 10\gamma$) to peak fields of 50, 100, 200, 300, 400 and 600 Oe. In several cases this range was extended up to about 1800 Oe and occasionally supplemented by demagnetization data from another specimen from the same sample. Typical intensity decay curves are shown in Fig. 2.

The median destructive fields (mdf) of the specimens were typically between 50 and 200 Oe (65 per cent of the specimens), whereas the mdf for 10 per cent of the specimens was below 50 Oe and for 10 per cent above 600 Oe. No correlation between the mdf and the intensity of NRM was found.

The major carriers of the NRM are titanomagnetite and the high coercivity components of the remanent magnetization are probably due to fine grained single or pseudo-single domain titanomagnetites, which may be present as the cumulus/intercumulus phase and/or as secondary alteration products of the mafic minerals. Two samples (11.21 and 12.21 from

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Table 1. Palaeomagnetic results.

Site	N	D_m (°)	I_m (°)	R	k	α_{95} (°)	Cleaning fields (Oe)	Strike ± 2° (°)	Dip ± 1° (°)	D_{corr} (°)	I_{corr} (°)	R	k	α_{95} (°)	Palaeomagnetic pole				
															p	Lat. (°)	Long. (°)	δp (°)	δm (°)
NW-component																			
1	2	343	−34	1.8030	5	44.0	0	90	20 S	345	−15								
2	8	309	−23	7.8986	69	6.7	400–600	89	20 S	313	−10								
3	6	322	−16	5.6719	72	7.9	200–1000	104	24 S	325	−3								
4	10	318	−22	9.8630	66	6.0	200–600	86	28 S	321	0								
5	6	319	−14	5.9311	73	7.9	400–1000	92	27 S	320	6								
6	10	329	−17	9.7177	32	8.7	400–1400	125	31 SW	334	−4								
7	11	327	−22	10.7906	48	6.7	600–1000	108	28 S	332	−4								
11	7	333	−15	6.9129	69	7.3	100–1000	188	24 W	340	−24								
Site mean:	8	325	−21	7.8592	50	7.9				329	−7	7.7862	33	9.8	94	19.4 N	224.7 E	5	10 dip corr.
Sample mean:	60	324	−20	57.9470	29	3.5									101	11.2 N	226.7 E	4	8 in situ
SW-component																			
1	10	243	7	9.7297	33	8.4	600–1830	90	20 S	242	−5								
2	3	232	24	2.9635	55	16.8	50–1870	89	20 S	228	12								
3	5	237	2	4.4178	7	31.5	0–1000	104	24 S	239	−15								
4	2	240	8	1.7933	5	45.0	50–300	86	28 S	240	−4								
5	2	253	−9	1.9803	51	13.9	600–1000	92	27 S	259	−17								
6	4	220	−2	3.7284	11	21.1	1830	125	31 SW	221	−33								
7	3	236	11	2.9816	109	7.8	0–1830	108	28 S	236	−11								
11	3	238	6	2.8476	13	22.3	0–600	188	24 W	237	−15								
Site mean:	8	237	6	7.8085	36	9.3				238	−11	7.7077	24	11.6	96	19.1 S	308.4 E	6	12 dip corr.
Sample mean:	32	238	6	29.8863	15	6.9									87	11.1 S	312.0 E	5	9 in situ

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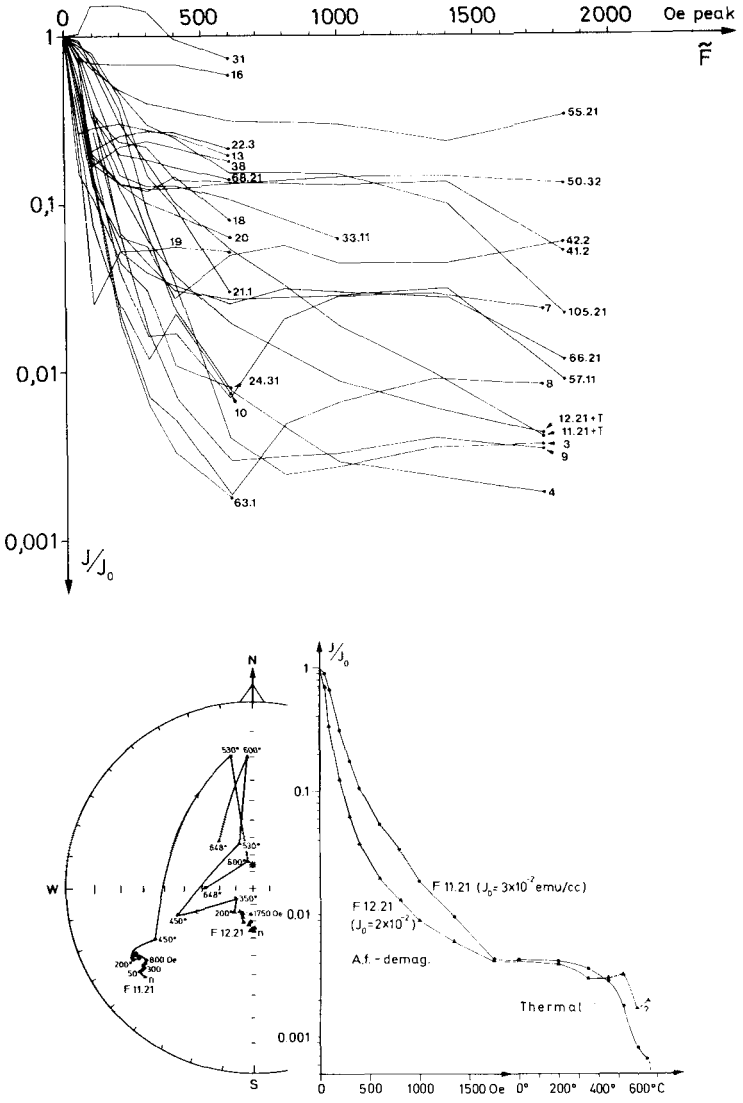


Figure 2. (a) Typical intensity decay curves by af demagnetizations. Numbers indicate specimen number, *cf.* Fig. 3. (b) Af – and thermal demagnetizations of two specimens very stable against af demagnetization. After migration towards the SW direction at moderate temperatures, a NW direction emerges at temperatures above 500°C.

site 2) were especially stable against af demagnetization with little change in initial remanence; they were therefore thermally demagnetized after af demagnetization to 1750 Oe peak. The resulting thermoremanent curves (Fig. 2b) indicate the presence of two high coercivity magnetic phases in the remaining RM (0.4 per cent of the NRM intensity only), one with Curie point at approximately 560°C (titanomagnetite) and one with Curie point above 650°C (haematite). Although these specimens are not typical for the gabbro as seen in Figs 3.1–3.6 they indicate that haematite may be present at least occasionally. The directional results of the demagnetization experiments of the gabbro specimens are shown in the stereograms of Fig. 3, and the resulting mean directions of characteristic remanent magnetic components with Fisher's (1953) statistics are summarized in Table 1.

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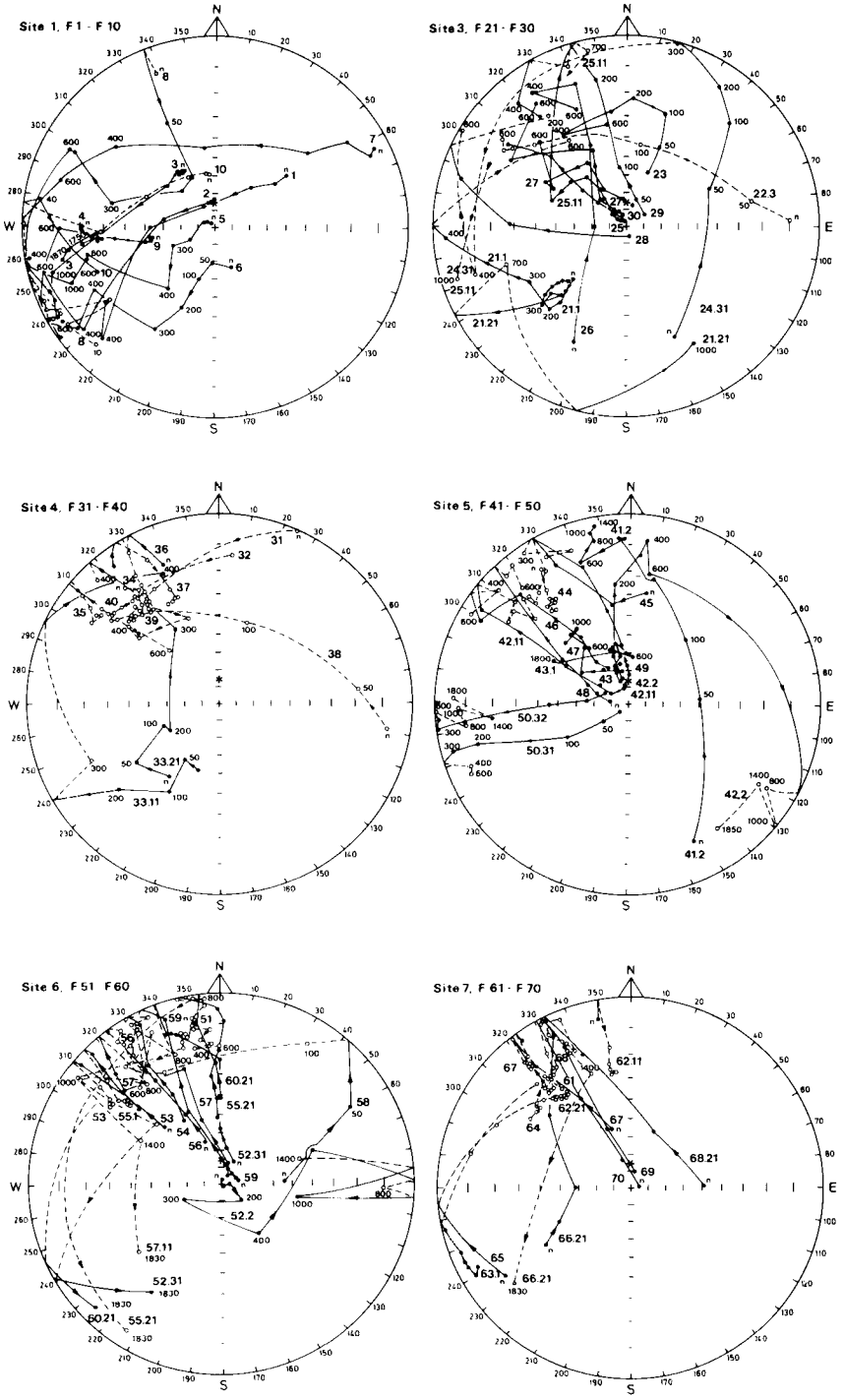


Figure 3. Change in direction of the remanent magnetization by af demagnetization for all specimens from six sites investigated. Small numbers indicate the peak magnetic field in Oersted, larger numbers indicate specimen number. Stereographic projection, dots are on lower hemisphere, circles on upper hemisphere. The SW direction is most obvious in site 1, the NW direction in the other sites. Asterisk indicates the present magnetic field direction.

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During the af demagnetizations (Fig. 3) the direction of the RM of the eight gabbro sites converged and in most cases stabilized in a characteristic up-dip NW-direction with a well defined site mean of $D = 325°$, $I = -21°$ ($\alpha_{95} = 8°$, $N = 8$) which becomes 329°, $-7°$ ($\alpha_{95} = 10°$, *cf.* Table 1) when applying dip correction (i.e. rotating the layering to a horizontal position).

Site 1, however, behaves in a different way. It converges against a subhorizontal SW-direction, and when taking a closer look at the demagnetization stereograms of the other gabbro sites, it appears that a characteristic SW-component is present in all the other sites, although not as clearly depicted as in site 1 which only sporadically shows the NW-direction (except specimens 8 and 10, Fig. 3).

The SW-direction is obviously not as well defined as the NW-direction, and a formal palaeomagnetic stability index such as that of Symons & Stupavsky (1974), which applies well for the NW-direction is of but little help in defining the SW-direction. In 17 cases the SW-component is revealed in or even dominates the NRM, and the remanence migrates to the NW during progressive af cleaning. In some cases the SW-component emerges in a moderate demagnetizing field between 200 and 1000 Oe, but stable high coercivity SW-components are only found in a few cases, notably in site 6. The coercivity spectra of the NW- and the SW-components apparently overlap, and both cover a wide range, probably reflecting the wide range in grain size of the magnetic phases present.

The best evidence for the relative magnetic chronology is seen in sites 3, 4 and 7 (Fig. 3), where the SW-component is clearly removed by cleaning in higher ac fields, thus illustrating the dominance of the high coercivity components of the NW-direction. In site 4 younger viscous components are nearly absent, making the picture even more clear. In this site, it also appears that the blocking period of the magnetic recordings was of sufficient duration to capture normal as well as reverse polarity of both directions.

As all the petrological events of relevance to the magnetic blocking of the remanence occurred below the respective Curie-points ($\leq 575°$C for titanomagnetite), the magnetizing processes must have occurred during the postmagmatic and metamorphic (and tectonic) history of the gabbro.

Further, the local temperature of the gabbro during the blocking of the SW-component was not higher than the blocking temperatures of the NW-component, otherwise the NW-direction could not have been preserved; both remanences appear to be carried essentially by titanomagnetites of equivalent Curie-points. The generally higher coercivities of the NW-component probably also indicate that at least part of the carriers of the NW-component are more fine-grained than those of the SE-component.

Finally, it should be noted that in site 1 only, five specimens (F1–5) have relatively high content of pyrrhotite as compared to magnetite; the Curie-point of pyrrhotite is fairly low (300–325°C), and site 1 is the only one in which the SW-component is completely dominant. This supports the other indications that the SW-component is younger than the NW-component.

Considering the change in direction from the SW- to the NW-quadrant shown by the thermally demagnetized specimens (Fig. 2), we conclude that the SW-direction was most likely acquired at lower temperatures (below about 450°C, Fig. 2) than the high coercivity NW-component, and is likely to be a younger magnetization, although high coercivity phases are also present.

Correlating the genesis of specific opaque phases with the two characteristic remanent directions, however, is not easy in spite of detailed information from polished sections and magnetic properties. No clear correlation with any specific phase or degree of oxidation has been noted, except the mentioned correlation at site 1 between pyrrhotite and the SW-

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component, and a tendency for the NW-component to correlate with high temperature secondary magnetite in the mafic phases together with the cloth-like microstructures of the ulvöspinel–magnetite intergrowth. This may account for part of the high coercivity components most likely due to single/pseudosingle domain grains, and may suggest that the high coercivity part of the NW-component was induced before the oxidation of ulvöspinel, the SW-component being still younger.

In Table 1 the site mean values of the SW-components present are summarized. The site mean of the SW-direction is found to be $D = 237°$, $I = 6°$ ($\alpha_{95} = 9°$, $N = 8$) without dip correction, and $D = 238°$, $I = -11°$ ($\alpha_{95} = 12°$) with dip correction applied. The precision parameter is generally lower for the SW- than for the NW-component.

4 Discussion of the palaeomagnetic data

In Fig. 4 the two mean directions of Table 1 are plotted in stereographic projection before and after dip correction, and it is seen that they are very nearly orthogonal to each other. Correction for dip does not improve the scatter of either direction; actually α_{95} is slightly increased (by about 12 per cent) in both cases. The change in mean direction is only 14 and 17° for the NW- and SW-directions respectively and only the inclination is essentially affected. The initial dip of the rhythmic layering before deformation is unknown; a qualified

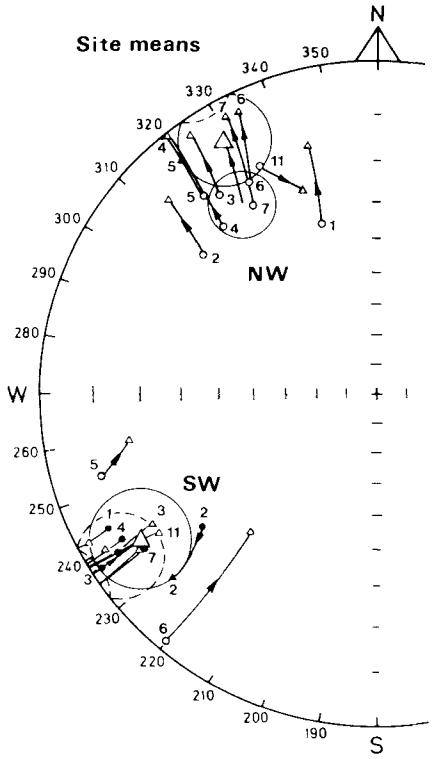


Figure 4. The two characteristic sets of remanent mean directions for each gabbro site, the NW and the SW components. Triangles are dip corrected directions, dots and circles are *in situ* directions (without dip correction). The circles of 95 per cent confidence are shown for the site means, α_{95} being slightly greater for the dip corrected ones.

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guess may be somewhere between 0 and 10° away from the margins of the intrusion, that is very roughly in the present down-dip direction for the sites shown in Fig. 1. The present dip of the sites (Table 1) varies between 20 and 30° and if dip correction must be applied, the true remanent mean values may be expected to fall somewhere between the dip corrected and non-dip corrected values. Whether or not the values should be corrected for dip will be discussed later.

The apparent palaeomagnetic pole positions P_1 and P_2, corresponding to the NW- and SW-direction, respectively, are 11° N, 227° E and 11° S, 312° E before dip correction; 19° N, 225° E and 19° S, 308° E after dip correction (Table 1); the geographical mean coordinates of the gabbro sites are 63.1° N, 11.6° E. In Fig. 5 the two poles, with and without dip correction, are plotted together with other palaeozoic poles of NW-Europe as compiled in Table 2.

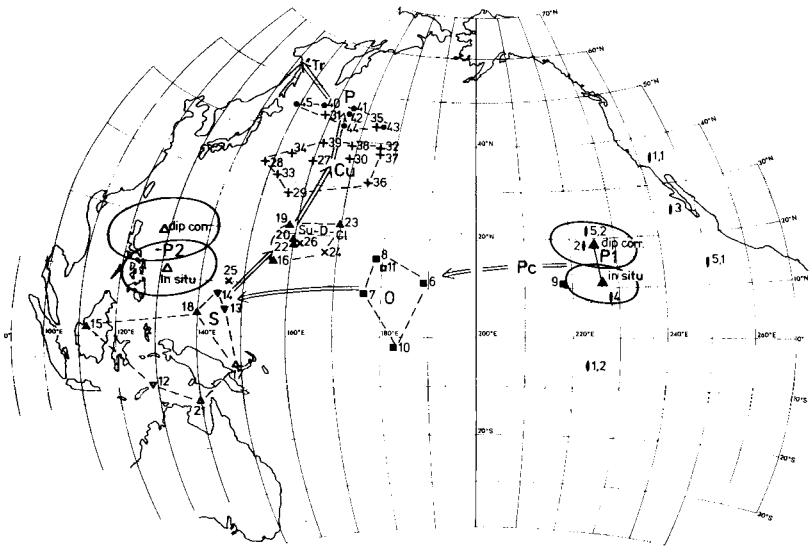


Figure 5. Apparent palaeomagnetic pole positions for NW-European Late-Precambrian and Palaeozoic poles as compiled in Table 2, the open symbols are reversed (Lambert zenithal equal area projection). The Late-Precambrian apparent polar wander curve as indicated with a broken arrow is not yet well defined, although a rapid shift is likely to have occurred somewhere in between Late Sveconorwegian (~800 Myr) and Lower Ordovician time (e.g. Abrahamsen 1977; Patchett & Bylund 1977). Other apparent shifts as indicated roughly by arrows are between the clusters of Ordovician–Silurian, Silurian–Devonian, Devonian–Upper Carboniferous and Upper Carboniferous–Permian poles. The data for Lower Palaeozoic poles for the British Islands are somewhat scattered. Significant experimental errors are unlikely; the scatter of results reflects polar movements during this interval, differential strike slip movements and differences on either side of the Iapetus suture. The apparent pole position P_1 of the NW direction of the Fongen Gabbro (with and without dip correction and with 95 per cent oval of confidence indicated) is situated between several Late-Precambrian poles, and may be interpreted either as a polar migration of some 50–60° with a duration of at least ~10^6 yr in Upper Silurian–Lower Devonian, or as a horizontal block rotation of some 60–90° anticlockwise (or some 90–120° clockwise) after the primary magnetic imprint of the NW direction. The apparent pole position $-P_2$ of the reversed SW direction, being some 10–20° and 25–30° away from the Silurian and Devonian clusterings of poles respectively, may accordingly be interpreted as due to a secondary magnetization, imposed during final cooling after the regional Caledonian metamorphism and nappe translations. In this case the *in situ* pole $-P_2$ (without dip correction) should apply, and again the deviation may either be interpreted in terms of a minor polar migration or possibly a clockwise rotation of some 10–20° after the magnetic imprint of the SW direction.

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Table 2. Selected palaeomagnetic poles for NW Europe.

Pole no.	Age (Myr)	Pole position Lat. (°)	Long. (°)	No, of sites	α_{95} (°)	Formation/location	Reference*
1.1†	Pc (990)	35 N	242 E	13	7	Lower Torridonian sandstone, Scotland (58° N, 6° W)	2; Stewart & Irving (1974)
1.2	Pc (805)	6	43	81	5	Upper Torridonian sandstone, Scotland (58° N, 6° W)	2
2‡	Pc ($\lesssim$6–700)	19	223	13	10	Dykes and baked contacts cutting Longmyndian sediments, England (53° N, 3° W)	Lomax & Briden (1977)
3	Pc-?	26	243	6	17	Precambrian–Lower Palaeozoic rocks, Czechoslovakia	Bucha (1965)
4†	Eoc	8	228	(16)	18	Tillite Series, E. Greenland	Bidgood & Harland (1961)
5.1	Pc	14	250	(32)	(4)	Listed diabase dike, Denmark (55°.1 N, 15°.1 E)	Abrahamsen (1977)
5.2	Pc	21	224	(6)	(14)	Bölshavn diabase dike, Denmark (55°.1 N, 15°.1 E)	Abrahamsen (1977)
6	01	11	189	32	9	Younger gabbros, Aberdeenshire, Scotland (57°.4 N, 2°.5 W)	1
7	01	9	176	27	5	North Borrowdale volcanics, England (54°.7 N, 3° W)	1
8	01	16	179	15	9	Built volcanics, Wales (52°.1 N, 3°.6 W)	1
9	01	11	218	17	20	Mweelrea ignimbrites, Eire (53°.6 N, 9°.7 W)	1; Morris *et al.* (1973)
10	Ou	2	2	6	11	Ashgilian age minor intrusions, Wales (52°.1 N, 33°.3 W)	1
11	Ou (>443 ± 16)	−14	0	6	28	Sulitjelma gabbro, N. Norway (67°.2 N, 15°.4 E)	Piper (1974)
12	Sl	8	309	5	14	Lavas, Somerset and Gloucestershire, England (51°.5 N, 2°.5 W)	Piper (1975b)
13	Su (415)	5	146	5	24	Garabal Hill–Glen Fyne Complex, Scotland (56°.3 N, 4°.8 W)	1
14	Su (418)	8	144	6	5	Arrochar Complex and aureole, Scotland (56°.2 N, 4°.8 W)	1
15	Su (− D1)	2	112	5	17	Lavas, Lower Old Red age, Lorne Pl., Scotland (57° N, 6° W)	1
16	(Su–D)	14	156	2	12	Vigehavn and Bölshavn diabase dikes, Denmark (55°.1 N, 15°.1 E)	Abrahamsen (1977)
17	Su–D1	6	329	11	11	Lavas, Lower Old Red age, Glencoe, Scotland (57° N, 5° W)	1
18	Su–D1	4	140	50	6	Lavas and baked seds., Lower Old Red age, Scotland (57° N, 3° W)	1
19	Su	21	159	10		Ringerike sandstone, S. Norway (60° N, 10° E)	2; Storetvedt *et al.* (1967)
20	(O−)D1	19	160	23	18	Foyers granite, Scotland (57°.2 N, 0°.5 W)	1
21	D1	11	320	11	15	Lavas, Lower Old Red age, Cheviot Hills, England (55°.3 N, 2° W)	1
22	D1	19	160	8	15	Röragen sandstone, Norway (62°.5 N, 11°.9 E)	2; Storetvedt & Gjellestad (1966)
23	Dm	22	170	(41)	9	Kvamshesten Old Red sandstone, SW Norway (61°.4 N, 5°.6 E)	2; Lie *et al.* (1969)
24	Cl	17	167	21		Kingshorn Lavas, Scotland (56° N, 3° W)	2
25	Cl	10	147	4		Derbyshire Lavas, England (53° N, 1°.5 W)	2

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26	Cl	18	161	17	8	Carboniferous Lavas, Scotland (56° N, 3° W)	2
27	C	34	160	7		Lancashire sediments, England (54° N, 3° W)	2
28	C	33	148	6		Derbyshire sediments, England (53° N, 1°.5 W)	2
29	Cu	28	157	10		Midland sills, England (52°.5 N, 2° W)	2
30	Cu (281)	37	169	34	~5	Whin sill, England (55° N, 2° W)	2
31	Cu (295)	44	159	5	5	Whin sill, England (55°.5 N, 1°.7 W)	2
32	Cu	39	177	76	4	Inner Sudetic Basin, Czechoslovakia (50°.6 N, 16°.1 E)	2
33	Cu	31	152	65	5	Plzen Basin sediments, Czechoslovakia (49°.8 N, 13°.3 E)	2
34	Cu	36	154	73	3	Kladuo–Rakovink Basin sediments, Czechoslovakia (50° N, 14° E)	2
35	Cu	43	174	8	14	Lower Silesian Volcanics, Poland (50°.5 N, 16°.5 E)	2
36	Cu (287)	31	174	9	2	Billingen dolerites, Sweden (58°.5 N, 14° E)	2; Priem *et al.* (1968); Mulder (1971)
37	Cu (– Pl) (279)	38	177	3	5	Hanneberg dolerites, Sweden (58°.5 N, 12°.5 E)	2; Priem *et al.* (1968); Mulder (1971)
38	Cu (– Pl) (283)	39	169	13	7	qz-dolerites, Skaane, Sweden (55°.5 N, 13°.5 E)	Bylund (1974); Klingspor (1976)
39	Cu (– Pl)	39	161	10	3	Ny-Hellesund diabase, S. Norway (58° N, 7°.8 E)	Halvorsen (1970)
40	Pl	47	157	27	1	Oslo igneous rocks, Norway (59°.7 N, 10°.4 E)	van Everdingen (1960); Heier & Compston (1969)
41	P	47	165	8	7	Rhombporphyric dikes, Bohuslän, SW Sweden (58°.7 N, 11°.5 E)	Thorning (1973); Thorning & Abrahamsen (1979)
42	Pl (280)	46	165	22	7	Exeter lavas, England (51° N, 4° W)	2
43	Pl	43	175	14	11	Lower Silesian volcanics, Poland (51° N, 15°.5 E)	2
44	Pl	43	165	11	8	Volcanics, Krakow, Poland (50° N, 20° E)	2
45	Pu	46	148	30	2	Dome de Barrot redbeds, France (44° N, 6°.8 E)	2
F1	Dl	19	225	8	8	Fongen gabbro, Norway (63°.1 N, 11°.6 E)	Present study
F1 §		11	227	8	10	Fongen gabbro, Norway (63°.1 N, 11°.6 E)	Present study
F2	Dl	–19	308	8	9	Fongen gabbro, Norway (63°.1 N, 11°.6 E)	Present study
F2 §		–11	312	8	12	Fongen gabbro, Norway (63°.1 N, 11°.6 E)	Present study

★ 1: Briden *et al.* (1973), Table 1. 2: McElhinny (1973), Appendix.

† Poles 1 and 4 are from the opposite side of the Iapetus suture from the others listed, and are therefore not strictly comparable.

‡ Pole 2 is now considered to be of Silurian age (Piper, private communication).

§ *In situ* (no dip correction).

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As a first approximation, the pole $-P_2$, corresponding to a reversed SW-direction, may readily be interpreted as a reversely polarized palaeomagnetic pole of Caledonian, Upper Silurian–Lower Devonian age, as it is not far away from other known poles of this age (see further discussion below).

The position of the pole P_1 of the NW-direction corresponding to the dominant NW magnetization of the gabbro is, however, not readily explained. It is situated about 90° away from the P_2 pole, and is placed in the NE-Pacific area (Fig. 5). The pole P_1 cannot be explained as a much younger magnetization because of the absence of any correlation with post-Palaeozoic poles, bearing in mind the dominant nature of the P_1 pole over the P_2 pole and their apparent relative age.

As components of both polarities appear to be present in some specimens (notably 22.3, 24.31, 31, 32, 38, 41.2, 42.2, 52.2, 58, 68.21, 105.21 and 107) it could be argued that the NW-direction is just the accidental mean of two shallow SW-subcomponents of opposite polarity (SW and NE) combining to a NW-direction, possibly supplied by a steep normal component of a recently induced viscous component. This is quite unlikely, because the convergence of the NW-direction is so well pronounced (it would be expected to show a random variation confined to the vertical plane orthogonal to the SW–NE axis) and because the recent viscous remanent magnetization, if not completely destroyed by the cleaning process, would tend to parallelize the combined directions in a steep northerly downdip (or southerly updip in case of a reversed field) direction in the vertical plane containing the SW-direction.

It would therefore appear that both poles must be explained in a Palaeozoic context. One possibility is that the P_1 pole could be an original Silurian/Devonian TCRM, the apparent palaeomagnetic field having made an excursion of about 90° to the pole P_1 in Late Caledonian time, which was stable long enough to cause the NW-direction to be locked in firmly at the expense of the (probably slightly younger) P_2 pole of the general Caledonian SW direction. Considering the dimensions of the gabbro, the depth of emplacement (15–20 km) and particularly the synorogenic environment, the total cooling time of the gabbro to below the active blocking range of the TRM, of which the thermomagnetic blocking range is a considerable part, must have been of the order of several millions of years (e.g. Jaeger 1968). This palaeomagnetic polar excursion explanation is not very plausible, as out of about 50 published palaeomagnetic investigations of NW-European Palaeozoic formations (e.g. Creer 1970; McElhinny 1973) there are only two poles in the vicinity of P_1. These are Nos. 3 and 9 in Table 2, both of which are too old and with a high scatter ($\alpha_{95} \simeq 20°$), the palaeomagnetic reliability of No. 3 in addition being questionable (McElhinny 1973, note p. 289). The Late Ordovician (Caradocian) probably non-axial dipole field, which appears to have been recorded by some hypabyssal intrusions in North Wales and in dyke swarms in northern England (Piper, private communication, 1978) is also too old for our case.

Another interesting possibility is that the two directions record the rotation of a block within the Scandinavian Caledonides. A block including the Fongen–Hyllingen complex must then be supposed to have rotated anticlockwise around a subvertical axis by an angle of about 90° relative to the normal NE or NNE updip Palaeozoic field direction after cooling down below the effective blocking temperatures (say *c.* 500–550°C) of the titanomagnetites present. This rotation could presumably have taken place during the nappe emplacements. During or shortly after the rotation, and after a reversal of the field polarity (both polarities are known from other investigations of Caledonian rocks), the SW-direction was then induced as a secondary TCRM, which now defines the 'Caldeonian' pole P_2 of the gabbro complex. As the $-P_2$ pole lies some 10–20° W of the main cluster of other Upper Silurian–Lower Devonian poles (Fig. 5), it could be postulated that this direction was

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locked in prior to another possible 10–20° final clockwise rotation before the end of nappe emplacement.

According to this hypothesis the dominant NW direction does not define a true palaeomagnetic pole but rather the local rotation of the pertinent block. Rotation in the opposite (clockwise) direction of about 90° is an equally possible alternative; in this case no reversal needs to have taken place between the two recording times.

It would be useful to try to relate the possible timing of the acquisition of the two magnetic components to the geological events in which the Fongen–Hyllingen complex has been involved. Contact metamorphic assemblages surrounding the complex indicate emplacement at about 5 Kb, equivalent to a depth of 15–20km, and intrusion was shortly followed by the peak of regional metamorphism at up to about 7 Kb and 650°C (Olesen *et al.* 1973). In view of the well defined nature of the NW-direction and the high temperatures involved during regional metamorphism, it seems likely that this magnetic component was established during cooling after the regional metamorphic peak. Broadly speaking the peak of regional metamorphism and deformation was followed by two phases of mesoscopic folding (Olesen *et al.* 1973), which appear to have had little effect on the competent gabbro complex. These were followed by a major fold phase, the Selbu–Tydal phase of Olesen *et al.* (1973) which influenced the shape of the gabbro complex (Wilson & Olesen 1975) and had the effect of steepening the dips of the rhythmic layering around Fongen mountain where most of the palaeomagnetic samples were collected. This Selgu–Tydal phase comprises essentially open folds with no associated axial plane foliation. The temperatures prevailing during this fold episode are poorly known, and it is not clear whether the NW and/or the SW components pre- or post-date this fold phase and should consequently be corrected for dip.

The closest and hence most relevant palaeomagnetic results from Scandinavia at present available are from the upper Ordovician synorogenic Sulitjelma Gabbro some 500 km to the NNE (pole 11 in Fig. 5), from the Upper Silurian Ringerike Sandstone 350 km to the south (pole 19), from the Lower or Middle Devonian sediments at Röragen 50 km to the south (pole 22), and from the Middle Devonian Kvamshesten Sandstone some 350 km to the south-west (pole 23).

The palaeomagnetic pole from the (older) Sulitjelma Gabbro, when uncorrected for dip, is in good agreement with those from the British Isles (poles 6–10), and the Sulitjelma pole clearly lies outside any block possibly involved in only a 'local' rotation in the southern Scandinavian Caledonides. The precision of pole 11, however, is quite low ($k = 6$, $\alpha_{95} = 28°$, $N = 6$), the sites sampled being metamorphosed, tectonized and folded (Piper 1974, 1975a; Mason 1975), and the constraints from this pole on possible rotational interpretations hence would be rather broad.

The Devonian Röragen Old Red Sandstones (pole 22) are late orogenic; deposited on rocks of the nappe which contains the Fongen–Hyllingen complex; they may therefore post-date the possible rotational movements of the block containing the gabbro. The palaeomagnetic poles for the Röragen Sandstone, the Ringerike Sandstone (pole 19) and the Kvamshesten Sandstone (pole 23) are mutually consistent and define an 'Old Red' Norwegian pole close to 20° N, 160° E. A moderate rotation of 15° anticlockwise for the Kvamshesten area was suggested by Storetvedt (1967), because of the unconformity and slightly younger age of the sediments as compared to the base, to make a better coincidence of the Norwegian Old Red poles. Considering the Fisher statistics of pole 23 ($\alpha_{95} = 9°$), however, a rotation of only a few degrees may suffice in this case. The poles of four supposed Palaeozoic dykes on the island of Bornholm (pole 16) are also close to this Devonian Old Red pole. The Röragen pole position therefore appears to have been established essentially after the suggested rotation of the pertinent block containing the Fongen–Hyllingen complex.

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5 Conclusions

In the Fongen–Hyllingen gabbro complex two characteristic palaeomagnetic directions are recorded, a well defined NW direction (P_1) and a less well defined but distinct SW direction (P_2), and it appears that the NW component predates the SW one. The $-P_2$ pole of the SW (reversed) direction lies fairly close to late Caledonian poles from Scandinavia. It seems that the P_1 pole was established during cooling after the peak of regional metamorphism and may record either a persistent geomagnetic excursion during Late Caledonian time, or a roughly 90° rotation of a crustal block in the Scandinavian Caledonides.

In the last decade numerous palaeomagnetic investigations have been published, suggesting azimuthal block rotations of crustal segments of local to regional scales in orogenic collision zones. In most cases the field geologist had no observations to corroborate such ideas prior to the palaeomagnetic evidence. Some minor or medium scale examples in the Mediterranean mobile belt are the Northern Calcareous Alps showing about 45° clockwise rotation with respect to Eurasia since the Jurassic (Mauritsch & Frisch 1978), the Corso-Sardinian Plate showing about 50° anticlockwise rotation during the Early Miocene (De Jong *et al.* 1973; Storetvedt & Markhus 1978), and the Iberian and Italian peninsulae also showing anticlockwise rotations of between 35 and 70° since the Triassic. These and other examples from Turkey and Japan are summarized by McElhinny (1973).

Probably the most detailed sampled Palaeozoic region is the British Isles, the older Palaeozoic palaeomagnetic data of which are known to be quite complex. During the Caledonian orogeny the isles appear to have behaved as a separate crustal plate (Piper 1978), and in a recent palaeocontinental reconstruction Morel & Irving (1978), although with some hesitation, suggest that between −450 and −400 Ma the British Isles (perhaps in part) rotated over 90° relative to Gondwanaland about a local axis. The Fongen–Hyllingen gabbro complex may have recorded the rotation of a comparable orogenic plate within the Scandinavian Caledonides.

Other apparently anomalous Palaeozoic palaeomagnetic results from Europe have been reported, such as two Irish palaeomagnetic directions from the Mid-Ordovician Mweelrea ignimbrites and the Connemara gabbros (Morris *et al.* 1973) which may be interpreted either as 40° anticlockwise and clockwise local rotations, respectively, or as geomagnetic excursions. It is hardly possible at this stage definitively to decide which hypothesis – geomagnetic excursion or block rotation – is to be preferred to account for the palaeomagnetic results from the Fongen–Hyllingen complex. Further palaeomagnetic investigations from the Scandinavian Caledonides, as well as structural evidence related to possible azimuthal rotation, may throw more light on this interesting problem.

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